A source-relative factor-two approximation for reverse-complement superstrings

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Abstract

Assuming the count, periodic-layer, and connection lemmas of the pinned ordinary shortest-common-superstring construction of OpenAI, we give a deterministic polynomial-bit-time factor-two algorithm for superstrings with reverse complements. The alphabet and its involution are explicitly encoded and may vary with the input; fixed letters, reverse-complement palindromes, empty requirements, duplicates, and containment are included. The extension uses one matched construction: a phase-free $2k+1$ count rule for self-reversed periodic texts, adjacent sorted profiles sharing both reflection axes, and an odd core of $2k-1$ layers in a $2k$-layer budget unit. Established connection and symmetric traversal operations complete the algorithm. The dependence is on the stated source's stronger structural lemmas, not merely on black-box existence of an ordinary factor-two algorithm. Sixteen finite and scalar Lean support lemmas cover specified interfaces only; the full algorithm is not formalized.

1 Statement, dependence, and antecedents

Let Σ be a finite alphabet with an involution $\mathrm{cm} : \Sigma \to \Sigma$. For a word $s = s_1 \cdots s_h$, put $\rho(s) = \mathrm{cm}(s_h) \cdots \mathrm{cm}(s_1)$ and $\rho(\varepsilon) = \varepsilon$. A word is feasible for a finite listed family $\mathcal{S}$ if, for every $s \in \mathcal{S}$, it contains s or $\rho(s)$ as a contiguous substring. Write $\mathrm{OPT}_{\mathrm{RC}}(\mathcal{S})$ for its minimum symbol length. Alphabet labels, the involution table, word symbols, and list boundaries are explicit; N denotes their total encoded bit length.

Theorem 1.1 (Source-relative approximation)**.** *Assume the structural results of [1] at revision* `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`*, with the contracts specified below. There is a deterministic algorithm, polynomial in N, which for every such $(\Sigma, \mathrm{cm}, \mathcal{S})$ returns a feasible word T with $|T| \le 2\,\mathrm{OPT}_{\mathrm{RC}}(\mathcal{S})$. The assertion includes arbitrary fixed letters and words, empty words and the empty list, duplicates, and substring redundancy. Neither the alphabet nor the word lengths are bounded constants.*

The source's truths are assumptions of this article. We do not audit them or derive an independent proof of the ordinary superstring theorem. In particular, an unspecified ordinary factor-two algorithm would not supply the contracts used here. Yamano–Shibuya's reduction in [2, Theorem 2] also explains why an unrestricted reverse-complement result cannot be advertised as bypassing the ordinary problem.

1.1 The precise source interface

The following labels identify results in the pinned source, not labels of this article. Its section files are under `build/sections/`.

1. `02-counts.tex`: the blocking functional, extension and ordinary period rules, and the occurrence and balance proofs `lem:counts`, `lem:base`. Their use requires a nonnegative count vector on a substring-closed set satisfying those rules. It does not require choosing the smallest ordinary count vector.

2. `03-layers.tex`: `lem:recorded`, `lem:layers`, `lem:matching-window`, `lem:agreement`, and `lem:max-rotation`. These provide recorded counts, sorted primitive passages, actual matching windows, and the two periodic comparison bounds.

3. `04-connections.tex`: `lem:ordered-windows` and `lem:band`; the *proof* of `lem:collective`; and `lem:host` after its hypotheses are established on a single host path. We state the relevant contracts in Section 4. The source collective statement has k layers and $K = kp$; our odd-core use has $2k-1$ layers and $K = kp$, and uses the proof's geometric inequalities.

4. `05-processing.tex`: its ordinary threshold, partition, and individual-link/request construction, applied to nonself text orbits.

5. `06-cycles.tex`: the proof of `lem:cycle-contact` and the free-period strategy of `lem:free-period` and `thm:connection`. We adapt the geometric proof to odd cores and justify the two-port quotient separately; the original one-sided connection theorem is not assumed equivariant.

6. `07-algorithm.tex`: finite record search, explicit coordinate representation, and the suffix invariant for spelling the output.

All these hypotheses and their symmetric applications are made explicit below. Source-paid arguments are recalled in our notation where needed to make the adaptation checkable.

1.2 Related operations and the credit boundary

The arbitrary-involution formulation, formal copies of fixed words, paired cycle selection, both-orientation periodic comparisons, and an 8/3 approximation occur in Yamano–Shibuya [2], especially Section 2, Lemmas 7, 9, 12, and Theorems 11–13. Their CPM paper [3] proves 3.75 and 2.875 guarantees and, in Lemma 18, compares an optimum with its entire reverse complement. Thus the generic use of T and $\rho(T)$ is an antecedent, not a contribution here. Fici et al. [4] already use distinct palindrome vertex copies and paired reverse paths in Sections 2–3, and reversal surgery in Lemma 6. Their Theorem 7 is a one-half *compression* guarantee; it is not a factor-two length guarantee.

Jiang, Li and Du studied superstrings with flipping, as described by Fici et al. and Yamano–Shibuya [5, 4, 3, 2]. The 1992 full text was unavailable in our bounded searches, so our comparison uses those later accounts. Fici et al. report a 4 length bound for a greedy-like reversal algorithm. The CPM 2026 account distinguishes the 4-approximate MGREEDY extension from a suggested 3-approximate TGREEDY extension; WABI 2026 summarizes the earlier bound as 3. We do not assign an independent theorem statement to the unread 1992 body.

Goldberg–Karzanov's symmetric decomposition and parity results [6, Theorem 3.1 and Corollary 3.2] precede the pairing operations below. Their Section 5 already exchanges complementary halves of two self-symmetric cycles meeting at two mate vertices. We credit that splice; our shared-axis lemma supplies the meeting hypothesis for possibly different ordered staircase profiles. Their setting has no fixed vertices, whereas our hierarchical graph has fixed vertices, including ε; we therefore give the needed parity and traversal proof explicitly. Bidirected incidence walks and balanced connected Euler criteria also appear in Medvedev et

al. [7, Section 2.2, Lemma 1] and Kundeti et al. [8, Section 6]. Their spectrum-constrained cyclic assembly results are not imported as exact algorithms for our arbitrary-length linear requirements.

Extremal and critical rotations are established tools in Breslauer–Jiang–Jiang [9, 10] and Mucha [11]. We cite the 1997 journal article as the archival reference, but our operational comparison uses the 1996 BRICS report RS-96-21; the journal body was not read, and report lemma numbers are not transferred to it. The report's concluding remarks credit Crochemore with a Lyndon minimum/maximum-rotation argument. The hierarchical representation and the separation between cycle-cover bounds and connectivity also precede this work; see the Collapsing framework [12, Sections 2.2 and 7.2].

The contribution claimed here is exactly one matched package: the phase-free self-text count rule in Lemma 2.1, the transfer to adjacent sorted profiles sharing both axes in Lemma 3.1, and the $2k-1$ odd core inside a $2k$-layer unit in Lemma 6.1. No separate novelty is claimed for phase parity, tagged counts, maxima or even rounding, decomposition, splicing, anchors, batch flips, request reclassification, rotations, cycle accounting, half-Euler extraction, or complexity bookkeeping.

1.3 Why the matched package is needed

For identity complement on $\{b, c\}$, take $\mathcal{S} = \{bbcbbc, bcbbcb, cbbcbb\}$. The ordinary rules symmetrized by orbit maxima and even rounding give one-letter counts $4, 2$, hence $W_{\mathrm{sym}} = 6$. But $\mathrm{OPT}_{\mathrm{RC}} = 7$: a length-six word cannot contain the two distinct length-six orientation orbits, and $bbcbbcb$ satisfies both. This refutes only a connection theorem that would retain those counts and yield output length at most the same W_{sym}. It does not refute factor two or all equivariant methods. The stronger rule below gives $W = 12$ on this instance.

A different diagnostic, $\{aaa, aab, abcaa, accc\}$ with $a \leftrightarrow b$ and c fixed, has optimum 10. Minimum constrained-cycle-cover orientation ties can lead to ordinary forced-count certificates 11 or 9. This shows that arbitrary minimum tie-breaking does not ensure the desired W certificate; it does not exclude all orientation-selection methods. These finite certificates motivate the construction and are not used to prove Theorem 1.1.

Boosting only fixed words is also insufficient for the intended proof: in $(abcacb)^{\mathbb{Z}}$, the layer $z(x) = x + 8$ has fixed-axis windows at starts $0, 3$, whereas its unique greatest forward rotation starts at 4. The two windows there are $cbabcac$ and $cbabcacb$, neither fixed. A count rule confined to fixed-axis starts would not justify the maximal-rotation cycle argument. The rule below holds at every phase of a self-reversed text.

2 Symmetric counts and their domination

Remove empty requirements, merge ρ-orbits, and discard an orbit contained in a different larger orbit. If s is contained in t, then $\rho(s)$ is contained in $\rho(t)$, so either orientation of the larger word satisfies an orientation of the smaller. This preprocessing preserves feasibility and the optimum. If nothing remains, return ε. Otherwise let R include both orientations of the surviving requirements, and let V consist of all their substrings, including ε. It is substring-closed and ρ-closed; required words are maximal in V. Let L be the total symbol length of one representative per surviving orbit, so $L \le N$ and $|V| = O(L^2)$.

2.1 Blocking and the recursion

A periodic text is a map $A : \mathbb{Z} \to \Sigma$ with $A(i+p) = A(i)$ for some positive p, not necessarily least. For $s = A[0:h)$ and $r \in V$, let $b_{A,s}(r)$ count the integers j such that

$$\begin{gathered} 1 \le j, \quad j+h \le |r|-1, \quad r[j:j+h) = s, \\ r(i) = A(i-j) \quad (1 \le i \le |r|-2), \qquad r(0) \ne A(-j), \quad r(|r|-1) \ne A(|r|-1-j). \end{gathered} \tag{1}$$

Put $B_A(s;m) = \sum_{r \in V \setminus \{\varepsilon\}} b_{A,s}(r) m(r)$. Only $|r| \ge |s|+2$ contributes. For an occurrence aligned with A in a finite word, the nearest mismatches on both sides, if present, determine a unique bracket. Call it blocked when that bracket belongs to V; absence of either mismatch means unblocked. Consequently $B_A(s;N_T)$ counts blocked occurrences exactly. This is the blocking contract of [1, Section 2]; its coefficients count placements, not just distinct bracket words.

Define $m(s)$ in decreasing length, simultaneously on each ρ-orbit. Missing words have count zero. The lower bounds are

$$m(s) \ge \sum_c m(cs), \qquad m(s) \ge \sum_c m(sc), \tag{2}$$

with required lower bound 1 for nonfixed words and 2 for fixed words. For $v = A[0:h)$, $w = A[0:h-kp)$ nonempty, $k \ge 1$, impose

$$\begin{aligned} &m(v) > B_A(v;m) \quad \Longrightarrow \quad m(w) \ge B_A(w;m) + g(A,k), \\ &g(A,k) = \begin{cases} 2k+1, & A \text{ self-reversed}, \\ k+1, & \text{otherwise}. \end{cases} \end{aligned} \tag{3}$$

Here self-reversed means that for some integer c, $\mathrm{cm}(A(j)) = A(c-1-j)$ for every j; equivalently its reverse complement is a translate of itself. Take the maximum of all bounds for both words of an orbit and assign it to both. At a fixed word round upward to even. Every bound refers only to longer words, so this defines an explicit recursion. Enumerate every $v \in V$, $1 \le p < |v|$, and k with $kp < |v|$, retaining a rule precisely when repetition of $v[0:p)$ agrees with v. This includes all nonleast periods and all phases represented by such words. No fixed-word-only $2k+2$ variant is used.

Lemma 2.1 (Tagged-text domination)**.** *For every feasible T and every nonempty $s \in V$,*

$$m(s) \le N(s) := N_T(s) + N_{\rho(T)}(s).$$

Thus $W := \sum_{a \in \Sigma} m(a) \le 2\,\mathrm{OPT}_{\mathrm{RC}}(\mathcal{S})$, and $m(s) \le 2L$.

Proof. Treat T and $\rho(T)$ as distinct *tagged components*, even if their word labels coincide. Then N is symmetric and even at every fixed word. It satisfies the required and extension bounds. Descending induction for the ordinary $k+1$ rule is the proof of [1, `lem:counts`] on this disjoint collection: an unblocked v gives $k+1$ period-spaced unblocked prefixes, or a positive longer-bracket deficit contributes that many placements. The blocking functional is linear over the components.

We prove the additional $2k+1$ bound. Translate so that $v = A[0:h)$, $|w| = h - kp$. In a self-reversed text, $\rho(v)$ occupies $A[c-h:c)$. The possible starts of a copy of w inside this interval form an integer interval of $kp+1$ integers. At least k of them have residue 0 modulo p, and hence spell w in the required A alignment. The original v contains $k+1$ such copies, at $0, p, \ldots, kp$.

If $N(v) > B_A(v;N)$, choose an unblocked v in one tagged component. All $k+1$ internal copies are unblocked: a mismatch bracket of an internal copy must enclose the entire matching

v interval. Reflect this occurrence into the other component. Each of the indicated k aligned copies in $\rho(v)$ is likewise unblocked. A bracket blocking one would enclose $\rho(v)$; its reflection is in V and would block the original v. End mismatches remain mismatches because cm is a bijection. Tagging makes the two sets of occurrences disjoint. Therefore

$$N(w) - B_A(w; N) \geq (k+1) + k = 2k + 1.$$

Otherwise $N(v) = B_A(v; N) \geq m(v) > B_A(v; m)$. Write $\Delta(r) = N(r) - m(r)$ for longer words. Induction makes these nonnegative. Some bracket word r has $b_{A,v}(r) > 0$ and $\Delta(r) > 0$. Its chosen placement contains $k+1$ counted placements of w. The reflected bracket in $\rho(r)$ contains at least k of the aligned shifted-phase placements just described; both endpoints remain mismatches, so they are counted by the same functional for w. If $r \neq \rho(r)$, symmetry gives $\Delta(\rho(r)) = \Delta(r) \geq 1$, and these two bracket terms contribute at least $2k+1$ to $B_A(w; N) - B_A(w; m)$. If $r = \rho(r)$, the positive integer $\Delta(r)$ is even, hence at least 2; the original $k+1$ placements alone give $2(k+1) \geq 2k+1$. There is no double counting of a fixed bracket. All other terms are nonnegative. This proves the new bound in both cases.

Orbit maxima stay below the symmetric N, and even rounding stays below its even values at fixed words. Induction follows. Summing letter counts gives $W \leq 2|T|$. Concatenating one surviving representative per orbit supplies a feasible word of length L, proving $m(s) \leq 2L$. □

2.2 The balanced graph and fixed vertices

For each nonempty s, the hierarchical graph has an up edge $\mathrm{pref}(s) \to s$ and a down edge $s \to \mathrm{suf}(s)$; prefix and suffix here delete the last and first letter respectively. Up cost is one, down cost zero. Give these edges multiplicities

$$u(s) = m(s) - \sum_c m(cs), \qquad d(s) = m(s) - \sum_c m(sc). \tag{4}$$

By the telescoping proof of [1, `lem:base`], this graph G is balanced, has W up and W down copies, and uses every required vertex. Indeed both degrees at nonempty s equal $m(s) - \sum_{c,d} m(csd)$, and the total multiplicities telescope to W.

Symmetry gives $u(s) = d(\rho(s))$. Pair copies to define an edge-reversing involution τ, mapping the up edge at s to the down edge at $\rho(s)$. There are no fixed edges. At a nonempty fixed s, the displayed degree is even: nonfixed extension words pair and fixed extension words have even counts. At the empty vertex,

$$\mathrm{outdeg}_G(\varepsilon) = \sum_a u(a) = \sum_{|s|=1} m(s) - \sum_{|s|=2} m(s) \tag{5}$$

is also even. It is not the total up cost W. Adding a closed walk C together with its reversed image τC preserves balance, symmetry, and even outdegree at fixed vertices, since the increment there is $\mathrm{outdeg}_C + \mathrm{indeg}_C = 2\,\mathrm{outdeg}_C$.

3 Paired walks, sorted layers, and shared axes

3.1 Paired decomposition and the exit convention

At each nonfixed vertex choose a matching from incoming to outgoing edge copies and force its reversed matching at the mate vertex. At a fixed vertex, pair outgoing copies by a fixed-point-free involution; identifying incoming copies with their reversals gives the transition matching. This is possible by Section 2.2. The resulting cycles come in distinct reversed pairs. For if reversal fixed one cyclic edge order, its reflection would fix an edge or exchange two

consecutive edges, yielding a fixed transition. Both are excluded. Fixed tie orders make this construction deterministic. It is an elementary adaptation of paired-decomposition principles, not a separate contribution.

Lift each closed walk to windows $[x, e)$ in the periodic text formed by its cyclic up letters. Up steps increase e and down steps increase x; a turn with P up steps advances by (P, P). A word occurrence is recorded if a visited window contains it, counted once per starting residue modulo P. The telescoping up-step argument of [1, `lem:recorded`] gives total recorded count $m(s)$, even when $|s| > P$.

Group texts up to translation, reduce to least period p, and sort primitive passages by normalized exits. More explicitly, if original walk j has turn $d_j p$ and exit function Z_j, sort the multiset

$$\{Z_j(x + rp) - rp : j,\ 0 \le r < d_j\}$$

at each x. Its order statistics are layer exits $z_i(x)$. We index higher layers first. Put

$$f_i(x) = z_i(x-1), \quad l_i(x) = z_i(x), \quad z_i(x-1) \le z_i(x), \quad z_i(x-1) \ge x, \quad z_i(x+p) = z_i(x) + p. \tag{6}$$

A layer visits exactly $[x, e)$ for $f_i(x) \le e \le l_i(x)$, then takes a down step. The source sorting proof preserves the phase-labelled edge multiset: down steps are determined by exits, and vertical-edge multiplicities by entry-minus-exit cumulative counts. Each layer has p up steps and receives an additional budget p; the budgets sum to W. Every layer window spells a word of V.

The convention for occurrences is crucial. At a specified start t, a layer records $[t, t + h)$ exactly when its *exit* $z_i(t) \ge t + h$. Thus sorting preserves phase-specific recorded counts, not merely aggregate counts of word labels. Its entries decide whether that recorded word is also an actual vertex at start t.

3.2 Text orbits and adjacent pairs

A nonself text orbit comprises two translation classes exchanged by ρ. Their sorted layers pair rank by rank. Choose the canonical text to be the greatest forward rotation among *both* orientations. A basic unit consists of one representative layer and its reflected mate, with budget $2p$.

For a self-reversed primitive text fix

$$\mathcal{R}(x, e) = (c - e, c - x), \qquad \mathrm{cm}(A(j)) = A(c - 1 - j). \tag{7}$$

There are an even number of primitive passages, inherited from distinct reversed walk pairs. Reflection preserves the order of staircase profiles: it sends the upper region of a path to the upper region of the reflected path. The ordered decomposition is uniquely determined by its exit multisets. Its phase-labelled multiset is invariant under $\mathcal{R}$, so every sorted profile is individually fixed under this reflection, modulo period translation. This does *not* imply even multiplicity of each profile.

Lemma 3.1 (Shared-axis adjacent pairs)**.** *Pair the sorted layers of a self text from the top in adjacent pairs. Each pair shares both reflection-axis vertices in a primitive turn, and is therefore already connected. The two profiles may be different. Removing full pairs preserves all fixed-phase even tails.*

Proof. For a nonempty interval $[t, t + h)$ with $2t + h \equiv c \pmod p$, reflection fixes its alignment. It pairs its recorded occurrences on the original distinct reversed walks without fixed occurrences. Consequently their number is even. The exit convention makes this number exactly

$$\#\{i : z_i(t) \ge t + h\}, \tag{8}$$

which sorting preserves. Since exits are ordered, every even upper tail consists of whole top-aligned adjacent pairs. Whenever the upper member visits this axis vertex, the lower member records it too; its entry is no higher than the upper entry, so it actually visits the same vertex. The two threshold indicators agree for every pair, and removing a full pair preserves evenness.

Along one primitive turn, $x+e$ increases by one at every step and by $2p$ overall. It therefore reaches both classes c and $c+p$ modulo $2p$, at actual vertices. These are the two fixed-axis vertices, where the preceding argument applies. If the vertex is empty, its upper entry is x, and the lower entry, lying between x and that entry, is also x; both visit ε. No occurrence count of the empty word is needed. When $p = 1$, every window is an axis window and the turn has exactly two window visits; the same argument includes this case. □

3.3 Spliced paths for hosting

Let D_1, D_2 be a self pair and let U, V be its two shared axes in cyclic order. Form E from $D_1[U, V]$ followed by $D_2[V, U + (p, p)]$, and form F from the complementary arcs. These are monotone period-p paths with p up steps each; their labelled union equals that of D_1, D_2. Reflection exchanges the half-period arcs, so $\mathcal{R}(E) = F$ at the level of windows and phase-labelled edges. Coincident profiles use distinct labels. This is not an assertion about the original base edge-copy involution after sorting. Later rootedness uses shared vertices, and mirrored added walks receive new paired edge copies. Keep the original sorted profiles for all band geometry; use E, F only as host paths.

The exchange-of-halves operation has the antecedent in [6, Section 5]. That operation assumes the two meetings. Lemma 3.1 is what establishes them for the ordered profiles here.

4 Geometric connection contracts

A piece is *rooted* when its underlying undirected support is connected to ε. All edge multisets are balanced, and all base edges remain, so touching one vertex of a layer connects its entire support. Incidental connectivity does not change budget marks.

A *record* is an actual window on a target layer, together with an alignment in a source text agreeing throughout that window. The matching-window contract [1, `lem:matching-window`] says that the number of blocked recorded occurrences relative to A is $B_A(s; m)$, and every unblocked occurrence is contained in such an actual matching window. The nearest-mismatch correspondence proves the first assertion. For the second, follow the layer from the first window containing the occurrence until its left mismatch is deleted; a right mismatch cannot enter first without recording its bracket.

We also use [1, `lem:agreement`]: distinct aligned texts of periods p, q agree on fewer than $p + q$ letters, and on fewer than p if both have period p. Alignment matters, even within one translation class. The maximal-rotation contract [1, `lem:max-rotation`] says that if t starts the greatest forward rotation of A and every rotation of a different aligned period-q text D is at most that word, they cannot agree on all q positions starting at t. These comparisons use encoded-symbol order and include opposite orientations as well as different phases.

Lemma 4.1 (Ordered windows and bands)**.** *For windows $[a, b), [c, d)$ in a common text, spelling words in V, with $a \le c$ and $b \le d$, there is a hierarchical walk of up cost $d - b$, with a contact of length at most $\max(0, b - c)$. An ordered list costs its total end increase. If the last window is a period translate of the first by K, it closes at cost K. For any ordered band of period-p layers with top first end H and bottom last end E at t, a nonnegative multiple K of p with $H - E \le K$ joins the band at cost at most K.*

Proof. These are [1, `lem:ordered-windows`, `lem:band`]. If $c \le b$, delete to $[c, b)$ and append to $[c, d)$, using suffixes and prefixes. If $c > b$, delete to empty, traverse each intervening letter

by an up/down pair, then build the second window. The active alphabet's individual letters lie in V. Thus all intermediate vertices are legal. For a band, if $E \geq H$ the layers already share $[t, E)$. Otherwise grow from $[t, E)$ to $[t, H)$, touching each vertical range, and continue to $[t + K, E + K)$. The end increase is K. □

Lemma 4.2 (Collective proof in geometric form)**.** *Let any positive number of ordered period-p layers have top first end $r_H = f_H(a)$ and bottom last end $r_L = l_L(a)$. Let $R = [a, b)$ be a record and K a nonnegative multiple of p. If*

$$r_H \leq b + K, \qquad r_H - r_L \leq K, \qquad r_L + K \geq b, \tag{9}$$

a cost-K closed walk based at R touches all these layers.

Proof. This is the proof of [1, `lem:collective`], with its numerical layer-count restriction removed, not a verbatim use of its statement. Set $Y = \max(b, r_H)$. The inequalities give $b \leq Y \leq b + K$ and $f_i(a) \leq Y \leq l_i(a) + K = l_i(a + K)$ for every layer. Choose its first start $x_i \in [a, a + K]$ reaching height Y. Then $[x_i, Y)$ is an actual window. Layer order makes these starts nondecreasing from top to bottom. Follow R, these windows, and $R + K$ using Lemma 4.1. No step used the number of layers. □

4.1 The individual trichotomy

For a period-p layer with $f_i(a) \leq b$ and a different-text record $R = [a, b)$ of target period q, the source's Section 5 gives three cases. If a vertex has length at most p, root it by a round trip through ε. Otherwise, if $l_i(a) + p \geq b$, choose an actual $[a + p, e')$ with $b \leq e' \leq b + p$ and follow

$$R, \quad [a + p, e'), \quad R + p. \tag{10}$$

This cost-p link has a contact of length at most $\max(0, b - a - p) \leq q$. We use the weak bound $\leq q$ in cycle accounting. If neither case holds, reserve budget p for the actual requester window $[a, e)$, $e = f_i(a)$, which satisfies

$$a + p < e \leq l_i(a) < b - p. \tag{11}$$

Since $b - a < p + q$ and now $b - a > 2p$, necessarily $q > p$. A request may spend the recipient's budget; a planned link does not.

Lemma 4.3 (Single-path host contract)**.** *Let a monotone period-q host path have available budget q. For each of a nonempty set of distinct child text groups choose one actual host record $R_i = [a_i, b_i)$, agreeing with the child text of period p_i. The child texts are distinct up to translation from one another and from the host text. Every requester at R_i is an actual child window $[a_i, e)$ satisfying $a_i + p_i < e \leq b_i - p_i$, with its budget p_i reserved. Then the host and all requesters can be rooted within those budgets.*

Proof. This is [1, `lem:host`]; we recall its constructive content because it specifies what a spliced host must satisfy. Choose a child j of minimum period among the requesting groups and order the records from R_j to its next copy on the host. Translations by multiples of q transport each entire child coordinate system, preserving its labels and budget. Put $T = a_j + q$. Then

$$a_j \leq a_i \leq T, \qquad b_j \leq b_i \leq b_j + q.$$

Successive records obey $b_i - a_l < p_i + p_l$: if they overlap this is the agreement bound for distinct child texts; if not it is immediate. The initial host excursion visits all requester windows at j, ordered by end, then R_j. At each later record visit the clamped contacts

$$Q_{i,e} = [\min(a_i + p_i, T), \max(e, b_j)). \tag{12}$$

They lie inside R_i. Before clamping, their starts and ends are ordered because $a_i + p_i < e_i \leq b_i - p_i < a_l + p_l < e_l$; clamping preserves order. Each end is less than $T + p_j$, using the overlap with the next j record. Return to the first requester at j translated by q, whose end exceeds $T + p_j$. Lemma 4.1 gives a cost-q closed excursion and, on the final transition, a contact of length less than p_j.

For a requester at $i \neq j$, write $Q_{i,e} = [s, h)$. Then $a_i \leq s \leq a_i + p_i$ and $e \leq h \leq e + p_i$. For the last bound, if $h = b_j$, minimum $p_j \leq p_i$ gives $b_j < a_i + p_j + p_i \leq a_i + 2p_i < e + p_i$. In the child text the ordered list $Q_{i,e}, [a_i + p_i, e + p_i), Q_{i,e} + p_i$ attaches the requester at cost p_i. The requesters at j were touched directly, so one unspent p_j budget roots the short contact. This proves the contract, including the case with no later records. □

5 Routing requests through paired hosts

Process primitive-text orbits in increasing least period, breaking ties by canonical text and stable identifiers. Requests will go strictly upward in period. All records refer to the fixed base-layer decomposition. A basic pair reserves its entire $2p$ budget, and processing never splits that account.

For a nonself recipient pair use its two exchanged sorted layers E, F. For a self pair use the spliced paths of Section 3.3, each with budget q and together representing the recipient's $2q$ account. If requests arrive, choose one original request as an anchor and name E so that its original record lies there. Keep its whole child-orbit batch unflipped. Every other child orbit also uses one record for all its requests. If that record lies only on F, reflect the whole batch, so its record lies on E. Distinct child orbits still give distinct child texts, all of period smaller than q.

5.1 Reclassification before invoking the host lemma

A reflected requester may be right-anchored. At its new record $R' = [a', b')$, its reflected actual window is $[u, b') \subseteq R'$. Monotonicity gives $f(a') \leq f(u) \leq b'$. Reapply Section 4.1 to that layer and R'. Its first case is vacuous here: the original case-three requester had no vertex of length at most p, and reflection preserves lengths. Its second case gives a link to R' which will be executed *after* the host is rooted. Its third case gives exactly the left-anchored inequality in Lemma 4.3. The unchanged anchor batch still supplies at least one genuine third-case request.

Apply Lemma 4.3 once on the actual monotone path E to the surviving third-case requests and mirror the entire construction on F. The one-record-per-child, distinct-text, left-anchor, nonempty, and available-budget hypotheses have all been re-established; this is not an invocation on a disconnected union of profiles. Choose the minimum-period child only among these surviving groups. One of its requester-pair budgets pays the two short root contacts; that pair was touched directly by the two host excursions. Total cost is at most $2q$ plus all surviving requester-pair budgets.

Both host paths are rooted. Their union contains all of the original recipient pair. Now execute the second-case attachments with their own $2p$ accounts. They are attachments to rooted material, not new outgoing plans at a noncanonical phase. Thus all original requester units become rooted. A self child pair is connected at its axes, so touching its selected member roots both; a nonself pair uses the mirrored construction for its other member. This establishes the routing without a new geometric host lemma or separate novelty claim.

5.2 Two roles of the baseline

Choose the highest recipient *pair* as baseline if there are any recipients. Its two members have already been rooted. The unchanged anchor gives an actual record S of length $< q + p < 2q$ on one member, where $p < q$ is a child's period. The short-record member and the upper sorted member used to bound the available region need not coincide.

Root every unspent pair below the baseline with its own $2q$ budget. If a layer has a vertex of length at most q, use that root excursion and its mirror. Otherwise, writing the short record as $S = [a, h)$, a lower layer has an actual $[a, e)$ with $q < e - a \le h - a < 2q$. The ordered list

$$[a, h), \quad [a + q, e + q), \quad [a + q, h + q)$$

costs q and attaches the lower layer; add its mirror. For a nonself orbit, if S lies on the partner layer, do this construction in that partner text group with the same-rank lower member and then reflect. For a self pair all lower pairs lie below whichever member carries S. The upper member of the baseline pair is used subsequently for band tests. These two uses of the baseline have distinct hypotheses and budgets.

6 Upper groups and the matched odd core

After incoming requests and lower pairs have been handled, the available upper region has intact pair boundaries. Its budgets are untouched. We now either root its pairs, reserve outgoing links, or send requests using one chosen record per child text orbit.

6.1 Nonself orbits: the ordinary procedure and its mirror

Use the canonical representative group. If it has n available layers, let t start its greatest forward rotation and $H = f_H(t)$. When $H - t \le np$, an excursion through $[t, H)$ roots them within np. When a baseline exists with $l_{\text{base}}(t) \ge H - np$, a band of width np attaches them. Mirror the chosen operation; the paired cost is at most $2np$.

In the remaining case $H > t + np$ and the baseline, if present, lies below $H - np$. Choose the first k with $\#\{i : l_i(t) \ge H - kp\} \le k$, counting the whole representative group. It stops by n. The preceding failed threshold gives the bottom of the top k layers last end at least $H - (k-1)p$; for $k = 1$ this is immediate. Translation to $(t-p, t]$ gives the wide band $\le kp$ at every start. The ordinary $k+1$ rule forces an unblocked occurrence of $w = A[t : H - kp)$ at a different full-text alignment, and hence a record $R = [a, b)$ with $a \le t$, $b \ge H - kp$, $|R| < p+q$. Only canonical-group layers can carry the identical alignment.

Set $K = kp$, $r_H = f_H(a)$, and let r_L be the bottom last end of that top block. If $r_L + K \ge b$, reserve its K budget for the collective link. Otherwise use the individual trichotomy on all upper layers. If the collective link is made, use the trichotomy on the layers below it. In either case their first ends at a are at most b: when the test fails $r_H \le r_L + K < b$, and below a successful block $l_i(t) < H - K \le b$. Mirror every root, link, and request with its paired layer budget. This is precisely the ordinary procedure of [1, Section 5], with paired accounting. A collective block and its mirror form a unit of budget $2kp$; band joining each component costs at most $2(k-1)p$, leaving $2p$. An individual pair has the same saving with $k = 1$. The canonical orientation does not change with host routing.

6.2 Self orbits: threshold and odd core

Suppose the available upper region has $2r$ layers, paired from the top. If $r = 0$ there is nothing to do. Take the greatest forward phase t and $H = f_H(t)$. If $H - t \le rp$, an excursion through

the top vertex and its mirror roots all upper layers for at most $2rp$. If a baseline exists with $l_{\text{base}}(t) \geq H - rp$, a width-$rp$ band and its mirror do the same. Otherwise

$$H > t + rp, \qquad l_{\text{base}}(t) < H - rp \quad \text{if a baseline exists.} \tag{13}$$

Lemma 6.1 (Odd core and outside record)**.** *Choose the first $k \geq 1$ such that*

$$\#\{i : l_i(t) \geq H - kp\} \leq 2k, \tag{14}$$

counting the whole self group. Then $k \leq r$, and the top $2k-1$ layers, called the core, satisfy

$$l_{\text{core}}(t) \geq H - (k-1)p, \qquad f_H(x) - l_{\text{core}}(x) \leq kp \quad (x \in \mathbb{Z}). \tag{15}$$

For $w = A[t : H - kp)$ there is an actual record $R = [a, b)$ on a different aligned text of least period q, with

$$a \leq t, \quad b \geq H - kp, \quad b - a < p + q. \tag{16}$$

Proof. By equation (13), no layer at or below the baseline qualifies at $k = r$, so the threshold stops by r. If $k > 1$, failure at $k-1$ means at least $2k-1$ layers qualify there, giving the first bound. If $k = 1$, the core is just the top layer, whose last end is at least H. For $x' \in (t-p, t]$, monotonicity and periodicity give $f_H(x') \leq H$ and $l_{\text{core}}(x') \geq l_{\text{core}}(t) - p$. Translation proves the second bound.

The actual top vertex $v = A[t : H)$ is unblocked in its own text, so its count-rule trigger holds. The self-text rule supplies at least $2k+1$ unblocked occurrences of w. At most $2k$ have the identical full-text alignment, by equation (14) and the exit convention: each primitive layer has just one start residue with that alignment. An occurrence outside it gives an actual matching record by Section 4. Agreement bounds give equation (16); if $q = p$ the stronger bound $b - a < p$ holds. The alternative alignment may be another phase of the same group. □

6.3 Completing a paired unit

Set $K = kp$, $r_H = f_H(a)$, and $r_L = l_{\text{core}}(a)$. Then equation (15) and equation (16) give $r_H - r_L \leq K$ and $r_H \leq H \leq b + K$. If $r_L + K \geq b$, apply Lemma 4.2 to the $2k-1$ core layers, at cost K, and add its reflection. The final, $2k$th layer shares the axes with core layer $2k-1$ by Lemma 3.1. The whole $2k$-layer unit is therefore connected to its target within its $2kp$ budget. Alternatively, the sharp core band in equation (15) joins the core at cost at most $(k-1)p$; adding its mirror and using the shared axes internally joins the full unit for at most $2(k-1)p$, freeing $2p$. This includes $k = 1$: its core is one layer and its mate already meets it.

If the collective test fails, $r_H \leq r_L + K < b$, so *all* available upper pairs, including the top k pairs, enter the individual procedure. If it succeeds, only pairs below the top $2k$ layers do so; their last ends at t are $< H - K \leq b$, hence their upper first ends at a are at most b. Apply Section 4.1 to each such upper member, then mirror. Roots and links use at most $2p$; otherwise reserve a $2p$ request to period $q > p$. A self pair's two members already meet, so touching the chosen member is sufficient. Every request from this child orbit uses the *one* record supplied above.

Proposition 6.2 (Disjoint paired budget partition)**.** *After processing all text orbits, each basic pair is marked rooted or belongs to exactly one outgoing unit. A unit targets one actual layer and its reflected partner, and reserves only its own paired budget. A unit with budget $2kp$ can be joined internally for at most $2(k-1)p$. All requests have been fulfilled.*

Proof. Incoming requests use disjoint reserved child budgets and the still available recipient-pair budget. The baseline step spends only remaining lower-pair budgets. Easy branches spend only available upper budgets; in the hard branch, the collective unit and individual

pairs partition them. Every request goes to a strictly larger period, so it is fulfilled later by Section 5. Only unexecuted links remain outgoing. The internal saving is the ordinary sharp-band saving for nonself blocks and Section 6.3 for self blocks. Whole-pair removal preserves fixed-phase parity and pair boundaries. No incidental intersection is used to reassign or spend another account. □

7 Opening quotient cycles symmetrically

Make one node for each outgoing unit. Its target pair belongs to another unit or is rooted, so every nonsink node has one outgoing arc. A nonself unit has two ports, its representative block and its reflected block; each is internally connectable. A self unit is internally connected once its core is joined, so its conceptual ports coincide. A planned link pair joins corresponding source and target ports, possibly exchanging them. A quotient path therefore lifts to paths reaching endpoint ports. A twisted quotient cycle need not lift to two disjoint cycles, and we do not assume that it does.

7.1 Selfloops and maximal contacts

For a quotient selfloop, source and target have equal period p and different aligned text, even if the target is the opposite orientation. Thus $|R| < p$. Omit both planned links, join the unit internally, and use the saved $2p$ to root R and $\rho(R)$ for $2|R| < 2p$. Both ports are rooted.

For a cycle of length at least two, choose a unit with greatest canonical forward text, breaking ties by stable IDs. Every forward rotation of *either* orientation of the target text is at most this chosen source. Let q be the target period.

Lemma 7.1 (Paired short contact)**.** *The selected link pair can be realized within its reserved budget with one contact of length at most q and its reflection.*

Proof. For an individual plan, equation (10) gives the bound $\le q$. For a collective plan consider its representative block in the nonself case, or its odd core in the self case. Write $K = kp$, with record $R = [a, b)$, distinguished start t, and $H = f_H(t)$. If a constituent already has a vertex of length at most q, the collective construction suffices. Otherwise all constituent vertices have length greater than q.

The maximal-rotation contract and the canonical comparison give $b - t < q$. Put $x = \min(t, a+K)$. If $x = t$ this already gives $b-x < q$; otherwise $b-x = b-a-K < p+q-K \le q$. Thus $[x, b)$ is a short suffix of R. Every constituent first end at x exceeds b, since its vertex length exceeds $q > b - x$, and is at most H. With $F = f_H(x)$ we have $b < F \le H \le b + K$. Follow the ordered list

$$[a, b), \quad [x, b), \quad [x, F), \quad [a + K, b + K). \tag{17}$$

Its cost is K; the growth at x touches all constituents, and the short suffix is the desired contact. Its prescribed windows are actual vertices or substrings of them. This is the geometric proof of [1, `lem:cycle-contact`], now applied to the odd core where appropriate, not an unqualified use of its original block statement. The final self-layer meets its core partner. Reflect the constructed walk to obtain the second contact within the paired budget. □

Theorem 7.2 (Paired connection budget)**.** *One can add paired closed walks of total up cost at most W to G so that the resulting graph D is connected to ε. It is balanced, τ-invariant, and has even outdegree at every fixed vertex. Its total up cost U is at most $2W$.*

Proof. For a cycle of length at least two retain the selected link pair of Lemma 7.1, but omit the *target unit's* outgoing pair. These are different links. Join that target internally, saving

$2q$, and use the saving to root both short contacts. The quotient cycle with the omitted edge removed is a path ending at the target. Every lifted component meets one of the retained contact pieces, which touch both target ports. Rootedness propagates along the path to every source port, whether the lift was twisted or untwisted. This also covers mixed self and nonself cycles.

Different quotient cycles have disjoint accounts. Execute all other links in increasing distance from rooted sinks or opened cycles. Each uses its source unit's reserved account. Together with the earlier rooting and request operations, this roots every layer. The accounts partition the original layer budgets, whose sum is W.

Every addition was a closed walk and its reversed partner, with fresh paired edge copies. Hence balance, symmetry, and fixed-vertex parity persist by Section 2.2. All base edges remain, so required vertices remain used, and total up cost is at most $W + W = 2W$. □

8 Half-Euler extraction and polynomial bit complexity

Lemma 8.1 (Half-Euler walk)**.** *Let a finite nonempty balanced directed multigraph have a reversing involution on vertices and edge copies, no fixed edge, connected used support, and a fixed root in that used support. If every fixed vertex has even outdegree, there is a directed closed walk from the root using exactly one edge from each involution pair. It can be constructed in polynomial time.*

Proof. Quotient vertices and edges by the involution, retaining both incidences of each edge, including loops. At a nonfixed vertex orbit choose a representative v. Label an incidence positive if its compatible lift is outgoing at v, and negative if incoming there. The numbers of positive and negative incidences equal $\mathrm{outdeg}(v)$ and $\mathrm{indeg}(v)$, so they agree. At a fixed vertex the quotient degree equals its directed outdegree and is even.

Pair opposite signs at each nonfixed orbit and pair arbitrary incidences at fixed vertices. The resulting transition system decomposes the quotient edges into closed alternating trails. These lift consistently: at a nonfixed orbit an arrival through a positive incidence puts the lift at the mate of v, where departure requires a negative incidence; the other case is reversed. At a fixed vertex either state is available. The final/initial paired incidences give closure. If several trails remain, connectedness supplies two sharing a vertex. Cross-switch two incidence pairs there, aligning opposite signs at a nonfixed orbit, to merge the trails. This preserves all local conditions. Repeating produces a quotient Euler trail, which lifts from the fixed root and closes there, selecting one copy from each orbit.

This is the signed-incidence Euler construction underlying the antecedents in Section 1.2, with fixed vertices handled explicitly. The general lemma allows quotient loops. Our hierarchical graph has none, because an edge changes word length and ρ preserves it. □

Proof of Theorem 1.1. The empty cases were handled before the construction. Otherwise apply Lemma 8.1 to D from Theorem 7.2, rooted at ε. Since D has U up and U down edges and τ exchanges them, it has U edge orbits. The selected closed walk has U edges; half are up, since word length returns to zero. It visits a representative of every used vertex orbit, and thus one orientation of every required word. Append the letter of each up step to the output. At every moment the current vertex is a suffix of the output so far: up appends a letter to both, and down deletes the vertex's first letter. This is the suffix invariant of [1, Section 7]. Therefore the word is feasible and

$$|T| = U/2 \le W \le 2\,\mathrm{OPT}_{\mathrm{RC}}(\mathcal{S}). \tag{18}$$

For completeness we detail the finite realization and bit costs. All finite choices use encoded-symbol order and stable IDs, including orbits of equal least period, tied canonical

maxima, transitions, records, host anchors, threshold choices, and trail splices. Unused alphabet symbols may be ignored after reading and checking the explicit alphabet and involution; keep every used symbol and its mate.

There are $O(L^2)$ substring vertices of length at most L, and all counts are at most $2L$. Enumerating v, p, k as in Section 2.1 gives a polynomial list (the coarse bound $O(L^4)$ suffices). Evaluate blocking by enumerating every bracket word and position. Self-reversal is tested by comparing a primitive period word with rotations of its reverse complement. These are polynomial operations on explicit strings and polynomial-bit integers.

The base graph has $2W \leq 4L$ edge copies. Paired transitions and walk lifts are finite scans. Least periods, normalized passages, and their exit order statistics use explicit comparisons and sorting. The total number of primitive phases is W, so storing one period of each z_i and its text is polynomial. Both axis crossings and the host splices are scans through these stored paths.

Equality or order comparison of two periodic texts needs at most the sum of their periods, by the agreement bound. Normalize record starts to a turn, translating child coordinates together with the record. All actual windows have length at most L, and the resulting integers have polynomial bit length. There are $2W$ actual window visits per turn across all layers. To find a different-alignment record for w, enumerate them and each placement of w within them, at most $2WL$ candidates. Align the placement to its prescribed occurrence interval; reject identical full texts or windows not matching the source throughout. The count and matching-window arguments prove that a candidate remains.

Threshold searches, the individual tests, batch reclassification, host record order, and functional-graph cycle detection all operate on polynomially many stored objects. Collective intersections scan at most $K \leq W$ start coordinates. Ordered-window paths are materialized by prefixes, suffixes, and single letters. The paired budget proof bounds the final graph by $2U \leq 4W \leq 8L$ edge copies, including all additions. The quotient traversal and output take polynomial time in this explicit graph. Comparisons and output of variable-width symbol labels are charged their encoded bit lengths; no unbounded-label unit-cost operation, random choice, or infinite-text oracle is used. This establishes the claimed deterministic polynomial bit complexity at the full scope. □

A Scope of formal support and validation

A companion Lean package, `RCS128 formal v1`, contains sixteen support theorems. It does not formalize Theorem 1.1. Its independent formal review concerns these exact finite or scalar statements, with standard axioms only. The correspondence is as follows; theorem names are in namespace `RCS128`.

Declarations	Mathematical interface and limitation
`even_card_of_reflection`, `even_phase_card`	Even cardinality under a supplied free involution, optionally preserving a phase predicate. The actual tagged occurrences and their reflection are not constructed in Lean.
`filter_eq_range_at_boundary`, `adjacent_pair_of_even_tail`, `adjacent_exit_threshold`, `reflection_sorted_pair`	Even upper tails force adjacent top-aligned indicators to agree. The exact equality between phase counts and exit counts, sortedness, and phase preservation are hypotheses.

Declarations	Mathematical interface and limitation
`even_card_erase_pair`, `even_phase_after_pair_removal`, `lower_pair_visits_axis`	Removal preserves parity when both distinct elements belong to the underlying set and agree on the phase; entry order upgrades a recorded threshold to a visit. Axis existence and profile geometry are external.
`odd_core_from_previous_failure`, `first_threshold_core`, `exists_record_outside_alignment`, `odd_core_budget`	Minimal-threshold arithmetic, the $2k+1$ versus $2k$ pigeonhole step, and $2(k-1)p + 2p = 2kp$. Actual records, bands, and the $k = 1$ geometric case are external.
`shifted_copy_fits`, `fixed_deficit_surplus`	Offset fitting and an even positive deficit giving at least $2k+1$. The reflected bracket construction and complete domination induction are external.
`final_length_bound`	Scalar composition of $2\|T\| = U$, $U \le 2W$, and $W \le 2\,\mathrm{OPT}_{\mathrm{RC}}$. All three substantive premises are supplied outside Lean.

A.1 What the support does not establish

The frozen package does not formalize words, reverse complements, blocking, hierarchical graphs, sorted staircase geometry, spliced hosts, quotient cycles, half-Euler extraction, or polynomial bit complexity. The unblocked and nonfixed-bracket arithmetic $(k+1)+k = 2k+1$ is not part of its frozen count support. Removing pairs, re-indexing the sorted list, and pairing the remainder is not formalized as a composed layer operation. In the removal statements, "absent pairs" means absent from the *phase*; both distinct elements must still belong to the underlying set. The extra $1 < k$ assumption in `odd_core_from_previous_failure` is harmless but unnecessary; the separate first-threshold theorem includes $k = 1$. Most declarations are elementary finite or scalar lemmas. None should be described as a machine proof of the geometric construction.

The seven frozen author control declarations are mostly numerical or finite-set witnesses, not formal hypothesis-sensitivity tests of those theorems. The independent review added fifty-nine declarations that supply the sensitivity evidence and positive controls. We attribute that evidence to the reviewer, not to the original seven controls.

The recorded review rebuilt the support and reviewer sources with Lean 4.34.1 and inspected printed types and axioms. It performed module-only kernel replay. Pinned dependency `.olean` files were trusted read-only; no fresh replay or rebuild of all mathlib dependencies was performed. Compilation alone does not certify statement identity: a weakened factor-four statement passes the original harness. Identity therefore also rests on external manifest hashes and the reviewer's type audit. The original harness has declaration/token-scan limitations, closed for these frozen sources by that audit and the module replay, not by a universal claim about the harness.

A.2 Diagnostics and author-side provenance

Finite count, axis, splice, and graph checks are diagnostics, not proofs. An independent proof reviewer also wrote an end-to-end executable model and tested hosts, cores, cycles, and output. Actual multi-child flipped batches occurred zero times in the recorded model runs; the routing argument in Section 5.1 rests on its mathematical proof, not on an assertion that testing covered that branch. Per-request shadow checks do not change that limitation.

The original frozen candidate, its exact scope, and all failed intermediate constructions are preserved outside this manuscript; no historical novelty statement is silently updated there.

B Attribution, reading limits, and rights

This is a source-relative article by Myeongjin Kang, with no affiliation stated. AI assistance is part of its author-side provenance. The parent GPT-6 Astra supplied substantial author-side work on the reflection count argument, odd-core rule, spliced-host routing, cycle accounting, baseline distinction, empty-vertex degree correction, and scoped Lean implementation. Author-side agents developed the construction and checks; GPT-6 Astra prepared this manuscript. Separate Opus 5.5 agents performed the proof, formal-support, and contribution reviews. These were independent agent roles, not human refereeing or author-side checks relabelled as independent review. Those review results are not publication, acceptance, or worldwide-priority certificates.

The pinned source's sections 1–7 and operative proofs were read for this manuscript. Primary comparison is bounded: the WABI formulation, lemmas and algorithm sections, the CPM introduction and Lemma 18, Goldberg–Karzanov's specified decomposition, parity and splice passages, and the cited bidirected and rotation passages were inspected. Fici et al.'s arXiv v1 extraction was read throughout. The accompanying reading ledger records exact ranges and distinguishes this author's reading from the parent's broader reading. The Jiang–Li–Du 1992 body and Breslauer–Jiang–Jiang 1997 journal body remain unread under the explicit citation choices in Section 1.2. We make no exhaustive literature or worldwide-priority claim.

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